

TEAM MEMBERS

Gavin McIlvaine, Sean Gilday, Jay Stebbins, Elan Cote, Nathaniel Loomis

INTRODUCTION

The database manages users' reading habits by storing user profiles, book details, and interaction histories. It securely handles user credentials and personal data, catalogs books by genre, author, and publisher, and records activities like reading sessions, ratings, collections, and social interactions. Development utilizes Java and SQL, with team collaboration via Discord.

METHODOLOGY

CONCEPTUAL MODELING

- Developed an Extended Entity-Relationship (EER) diagram.

CONSTRUCT REDUCTION TABLES :

- Defined primary and foreign keys for relational integrity to construct reduction tables, ensuring efficient and accurate capturing of data.

IMPLEMENTATION OF SQL STATEMENTS

- Executed SQL Data Definition Language (DDL) statements to create structured relational tables with appropriate constraints and data types.

DATA GENERATION

- Used Mockaroo to create realistic and diverse datasets, populating entity tables with valid data.
- Generated relationship datasets through randomized data methods to ensure comprehensive coverage.

HYPOTHESES

- Books with less pages will have a higher average rating.
- During the summer months, books with more authors will have a higher average rating.

SYSTEM ARCHITECTURE

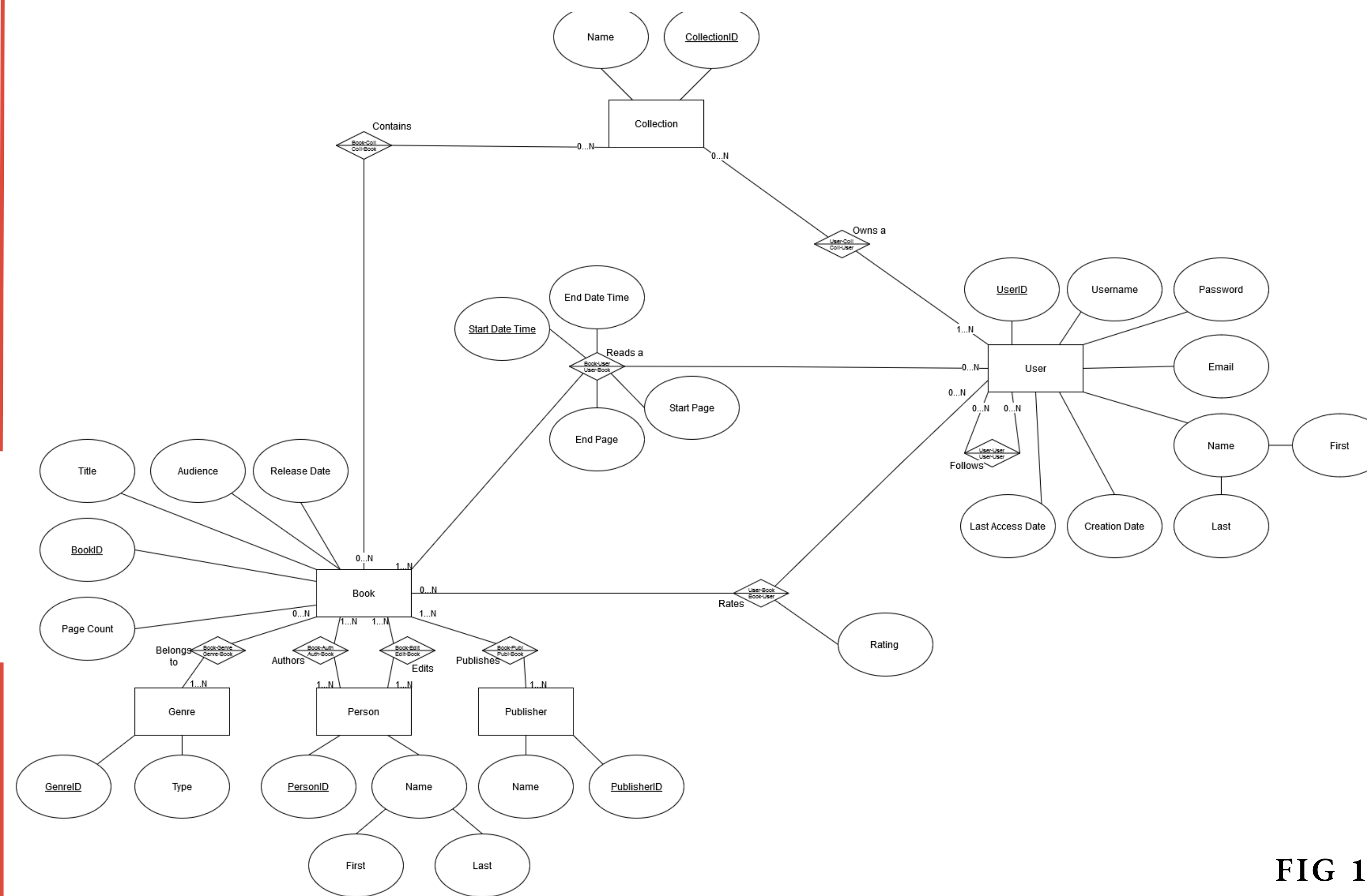


FIG 1.

ANALYSIS

Average Rating vs. Book Length

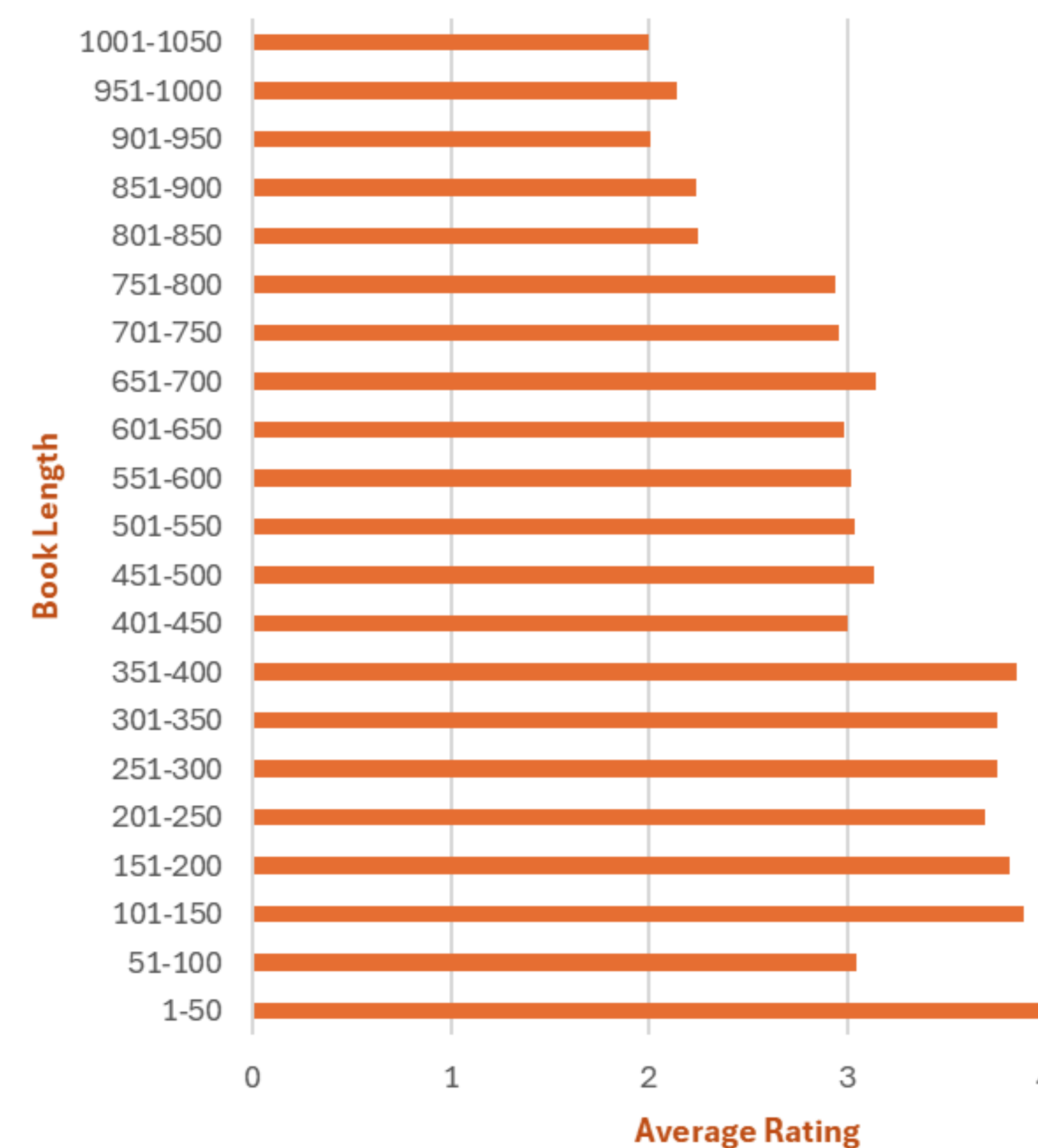


FIG 2.

For our first hypothesis we took the collection of our books and placed them into groups of 50 measured by page length. We then took these groups of 50 and calculated the average of each book's rating.

IMPLEMENTATION

For the overall implementation of the database, we used the relational model to help make our tables. SERIAL type for IDs were used as these are automatically increasing and unique per entry.

USER TABLE

```
CREATE TABLE "users"
( User_ID SERIAL NOT NULL PRIMARY KEY,
  Username VARCHAR(255) NOT NULL UNIQUE,
  Password VARCHAR(255) NOT NULL, First_Name
  VARCHAR(255) NOT NULL, Last_Name
  VARCHAR(255) NOT NULL, Email VARCHAR(255)
  NOT NULL, Creation_Date TIMESTAMP NOT
  NULL, Last_Access_Date TIMESTAMP NOT NULL,
  CHECK (Last_Access_Date >= Creation_Date) );
```

RESULTS/FINDINGS

- The data for the first hypothesis is segmented into groups of book lengths which show the average rating increasing as book length decreases, which proves the hypothesis. However, there is not an even number of books in each group, which may lead to the data being skewed.
- The data for the second hypothesis shows no increase in average rating between groups of author counts, which disproves the hypothesis.

NUMBER OF AUTHORS VS. AVG. RATING



For our second hypothesis, we queried all books grouped by the number of authors. We then filtered ratings to include only those from the summer months (June–August) and calculated the average rating for each author count group.